BLACK PEN Answer all questions first. **GREEN PEN** Then: tick correct answers; rewrite mistakes.

Combining Programming Constructs

Name _____ **Class** _____ **Date** _____ **Time** _____ **Teacher** _____

Do Now - Four retrieval questions

1. Why must a trace table follow executed assignments in order?

2. Which type suits a count of pupils?
   A) Real
   B) Boolean
   C) Character
   D) Integer

3. How do character and string data differ?

4. A limit of 30 appears five times in code. One copy changes to 32; the rest are missed. How would a named constant help? How does it differ from a changing score?

Today's objectives

Combine sequence, selection and iteration to solve problems. Apply the programming techniques to a new problem.

Key vocabulary - Read, cover, recall

Sequence
Initialise variables and read inputs before using their values.

Selection
Choose a route after evaluating a Boolean condition.

Iteration
Repeat a block using a loop where a task demands it.

Combined constructs
Real programs often need sequential code, decisions and repetition together.

BLACK PEN: Write one definition in your own words.

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Learn and practise

New Learning - Key ideas

Sequence
Initialise variables and read inputs before using their values. For example, consider a number guessing game.

Selection
Choose a route after evaluating a Boolean condition. For example, consider a quiz scorer.

Iteration
Repeat a block using a loop where a task demands it. For example, consider a book reminder.

Combined constructs
Real programs often need sequential code, decisions and repetition together. For example, consider a class rewards.

I Do - Worked model

A number guessing game is being developed. Explain how sequence and selection are used. [4 marks]

- Identify where this particular program needs control or values.
- Sequence [1]; Initialise variables and read inputs before using their values. [1].
- Selection [1]; Choose a route after evaluating a Boolean condition. [1].
- Give one accurate point and a linked reason for each technique.

We Do - Guided practice

1. A quiz scorer is being programmed. Explain why sequence matters. [2 marks]

2. For a book reminder, explain how selection is used. [2 marks]

3. A class rewards needs iteration and combined constructs. Explain the roles of both. [4 marks]

BLACK PEN Answer all questions first. **GREEN PEN** Then: tick correct answers; rewrite mistakes.

Think and write

Code First - Predict > Write > Test > Improve

YOUR TASK: Complete the condition to print only the even device IDs.

```
for device = 1 to 4
if ... then
print(device)
endif
next device
```

You Do - Independent exam questions

Q2. State TWO concepts needed for this programming technique. [2 marks]

Q3. Explain how selection and iteration can help a repeat ordering. [4 marks]

Q4. Explain the roles of sequence, iteration and combined constructs when creating a payment kiosk. [6 marks]

Exit Ticket - True or False?

Variables should be initialised and required inputs read before their values are used.

TRUE [ ] **FALSE [ ]**

BLACK PEN Answer all questions first. **GREEN PEN** Then: tick correct answers; rewrite mistakes.

Review & improve

GREEN PEN: Tick each correct point; correct mistakes.

Do Now - Model answers

1. Each variable value depends on the state produced by earlier executed instructions.

2. D - Integer

3. A character represents one character; a string represents a sequence of characters.

4. Define one named constant, such as MAX_CLASS_SIZE, and use it throughout. Change its definition once to update the limit. Use a variable for a score because it changes while the program runs.

I Do - Explain the model

Identify where this particular program needs control or values.

Sequence [1]; Initialise variables and read inputs before using their values. [1].

Selection [1]; Choose a route after evaluating a Boolean condition. [1].

Give one accurate point and a linked reason for each technique.

We Do - Model answers

1. Sequence [1]; Initialise variables and read inputs before using their values. [1].

2. Selection [1]; Choose a route after evaluating a Boolean condition. [1].

3. Iteration [1]; Repeat a block using a loop where a task demands it. [1]. Combined constructs [1]; Real programs often need sequential code, decisions and repetition together. [1].

Code First - Answer

```
for device = 1 to 4
if device MOD 2 == 0 then
print(device)
endif
next device
```

Expected output: 2
4

You Do - Self-mark

Q2: Sequence [1]; Iteration [1].

Q3: Selection [1]; Choose a route after evaluating a Boolean condition. [1]. Iteration [1]; Repeat a block using a loop where a task demands it. [1].

Q4: Sequence [1]; Initialise variables and read inputs before using their values. [1]. Iteration [1]; Repeat a block using a loop where a task demands it. [1]. Combined constructs [1]; Real programs often need sequential code, decisions and repetition together. [1].

GREEN PEN: Choose the hardest answer. Correct it on lined paper.